

Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Solution: Pre-Tutorial Assignment - 10

Date: Nov 2, 2023

Marks: 2

1. For the emitter-bias network in the active region (base-emitter forward bias, base-collector reverse bias) Fig.1. Let $V_{CC} = +20V$, $\beta = 50$, $R_B = 430k\Omega$, $R_C = 2k\Omega$, $R_E = 1k\Omega$. Determine the following (assume $V_{BE} = 0.7V$):

- (a) I_B
- (b) I_C
- (c) V_{CE}

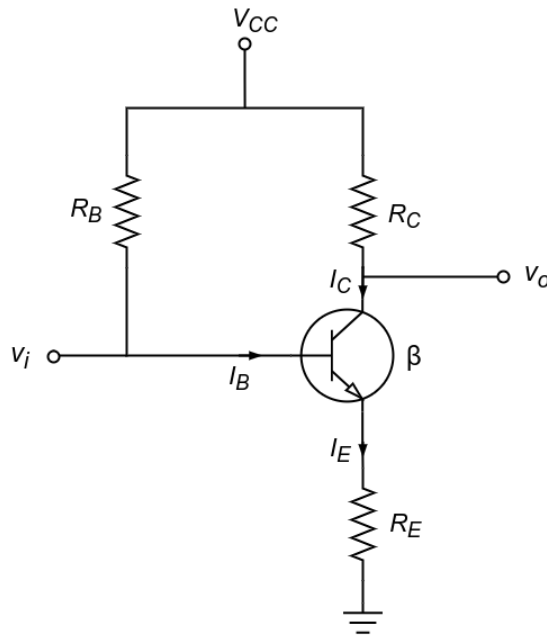


Figure 1:

Solution :

(a)

$$\begin{aligned} I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \\ &= \frac{20 - 0.7}{430k\Omega + (51)(1k\Omega)} \\ &= \frac{19.3}{481} \\ &= 40.12\mu A \end{aligned}$$

(b)

$$\begin{aligned}I_C &= \beta I_B \\&= (50)(40.12\mu A) \\&= 2.01mA\end{aligned}$$

(c)

$$\begin{aligned}I_E &= (\beta + 1)I_B \\&= (51)(40.12\mu A) \\&= 2.04mA\end{aligned}$$

$$\begin{aligned}V_{CE} &= V_{CC} - I_C R_C - I_E R_E \\&= 20V - (2.01mA * 2k\Omega) - (2.04mA * 1k\Omega) \\&= 20V - 6.06V \\&= 13.94V\end{aligned}$$